C9.1 About Exclusive accesses

The principles of Exclusive accesses are that a master performing an Exclusive sequence does the following:

- Performs an Exclusive Load from a location.
- Calculates a value to store to that location.
- Performs an Exclusive Store to the location.
  - The Exclusive Store fails if another master has performed a store to the location since the Exclusive Load. In this case, the store does not occur and the master does not change the value held at the location.
  - The Exclusive Store can pass if no other master has performed a store to the location since the Exclusive Load. In this case, the store can occur and the master can change the value held at the location.

Note

An Exclusive Load by a processor is caused by the execution of an instruction such as `LDREX`. An Exclusive Store by a processor is caused by the execution of an instruction such as `STREX`.

In the ACE protocol correct execution of an Exclusive sequence places requirements on both the master performing the Exclusive sequence and the interconnect.

For non-shareable locations the behavior is identical to that specified in AXI.

For shareable locations, the following requirements apply:

- The master is responsible for ensuring that it only updates the location if no other master can have performed a store to the location since the master performed the Exclusive Load. The term *master exclusive monitor* describes the monitor that must exist within the master component to meet this requirement.
- The interconnect is responsible for ensuring that if two masters attempt an Exclusive Store transaction to the same location and it is possible that the second master will have its copy of the location invalidated before its Exclusive Store transaction completes, then the interconnect must fail the Exclusive Store transaction from the second master. The term *PoS exclusive monitor* describes the monitor that must exist within the interconnect, at the point of serialization, to meet this requirement.

The term *Exclusive Store* is used to describe the action of a master executing an appropriate program instruction. An Exclusive Store can pass or fail and this result is known to the executing processor. When an Exclusive Store passes, this indicates an update to the data value at the address location. When an Exclusive Store fails, this indicates that the store has not changed the data value at the address location, and the Exclusive sequence must be restarted.

The term *Exclusive Store transaction* is used to describe the transaction issued on the ACE interface of a master. Not every Exclusive Store requires an Exclusive Store transaction. An Exclusive Store transaction can pass or fail and this result is made known to the master using the transaction response. When an Exclusive Store transaction passes, this indicates that the transaction has been propagated to other masters, but it does not indicate whether the Exclusive Store passes or fails. When an Exclusive Store transaction fails, this indicates that the transaction has not been propagated to other masters and therefore the associated Exclusive Store cannot pass.

All masters that attempt an Exclusive access to the same location must be using the same shareability for the location. If the location for the Exclusive access is shareable then all masters must be able to participate in the coherency protocol.

When first obtaining a copy of the exclusive location it is important that the line is not removed from another cache that is also performing an Exclusive sequence to the same cache line. For this reason, ReadClean or ReadShared must be used rather than ReadNotSharedDirty or ReadUnique.

A Load Exclusive, Store Exclusive sequence (`LDREX`, `STREX`) must use an Exclusive sequence. An atomic update, such as a swap operation or a read-modify-write atomic operation, is not required to use an Exclusive sequence. For such atomic updates, it is permitted to use a ReadUnique transaction which is not marked as Exclusive, if it can be guaranteed to successfully complete the atomic update with no other external action.

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